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FEDERAL SPECIFICATION

TABLE TOP ASSEMBLIES (PLASTIC THERMOSETTING RESIN LAMINATE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers table top assemblies of a decorative thermosetting laminate top attached to a plywood core with a paper phenolic laminate on the under surface. The edge of each table top assembly shall be covered with a metal or plastic trim.

1.2 Classification.

1.2.1 *Types.*—Table top assemblies covered by this specification shall be of the following types, as specified, according to the type of adhesive used in the plywood and in attaching the faces to the plywood core (see 6.2):

Type I.—Resorcinol or phenolic resins only.

Type II.—Urea resin only, or any combination of urea resin with resorcinol or phenolic resins.

1.2.2 *Grades.*—The decorative plastic laminate top shall be of the following grades, as specified (see 6.2):

Grade 1.—Standard

Grade 2.—Cigarette-proof

1.2.3 *Colors and patterns.*—The colors and patterns shall be as specified (see 6.2).

1.2.4 *Finish.*—The finish of the top surface of the decorative laminate shall be as specified (see 6.2).

1.2.5 *Trim.*—The composition and color of the trim shall be as specified (see 6.2).

1.2.6 *Sizes.*—The sizes of the table top assemblies shall be as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

L–P–406—Plastics, Organic: General Specifications, Test Methods.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

203440–53–(334)

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Uniform Freight Classification Rules.

(Application for copies of the publication referenced above should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

2.3 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Material in the decorative laminate.*—The decorative laminate shall be manufactured of paper and thermosetting resins. The cigarette-proof decorative laminate (grade 2) shall include an aluminum sheet.

3.1.2 *Material in the plywood core.*—The plywood core shall contain at least seven piles and shall be at least 0.75 inch thick, unless otherwise specified. The plywood shall have a face of close-grained-wood. The species and grade of wood shall be as specified in the contract or purchase order.

3.1.2.1 *Type I.*—For type I table top assemblies, the plywood shall be exterior grade (resorcinol or phenolic resin-bonded).

3.1.2.2 *Type II.*—For type II table top assemblies, the plywood shall be either water-resistant (urea resin-bonded) or any combination of urea resin-bonded with resorcinol or phenolic resin-bonded plywood.

3.1.3 *Laminated material on the back of the assembly.*—The backing shall be manufactured of paper and thermosetting resins and shall be at least 0.030 inch thick. It shall meet the following requirements in flexural strength and water absorption when tested as specified in 4.3.3.3.1 and 4.3.3.3.2, respectively.

Flexural strength, minimum, p.s.i.	12,000
Water absorption, maximum, percent	14

3.2 Decorative laminate.

3.2.1 *General requirements.* — The laminated thermosetting decorative material shall be uniform in quality and free from blisters, wrinkles or cracks. The decorative surface shall be free from small defects such as scratches, dents, and heat marks. The material shall be capable of being drilled, countersunk, tapped, sawed, and machined without chipping or delamination.

3.2.2 *Thickness.*—The decorative laminate shall be 0.062 ± 0.005 inch thick. If the decorative laminate is made of two layers with different resins and reinforcements in each layer, the top layer shall be at least 0.010 inch thick (see 4.3.3.1.1).

3.2.3 *Heat-conducting layer.*—Grade 2, decorative laminate shall contain a sheet of aluminum at least 0.002 inch thick. The distance of this sheet of aluminum from the top surface shall be between 0.010 and 0.020 inch.

3.2.4 *Colorfastness to light.*—There shall be no crazing and not more than a slight color change in the decorative laminate when tested as specified in 4.3.3.1.2.

3.2.5 *Flexural properties.* — The flexural strength shall be at least 15,000 pounds per square inch, the flexural modulus of elasticity shall be at least 800,000 pounds per square inch, and the deflection at failure shall be at least 0.020 inch when tested as specified in 4.3.3.1.3.

3.2.6 *Abrasion.*—The average weight loss per 1000 revolutions shall be less than 0.035 gram when tested as specified in 4.3.3.1.4.

3.2.7 *Water absorption.* — The water absorption determined as specified in 4.3.3.1.5 shall be less than 5.0 percent.

3.2.8 *Wetting and drying.*—The decorative laminate shall show no cracks or blisters in the top surface when tested as specified in 4.3.3.2.2.1 for type I or 4.3.3.2.2.2 for type II.

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3.2.9 *Chemical resistance.* — No swelling, blistering, or discoloration shall occur in the decorative laminate after 24 hours' exposure to the chemicals specified in 4.3.3.1.6.

3.3 Table top assembly.

3.3.1 *Impact resistance.*—When tested as specified in 4.3.3.2.1, not more than one of the ten test points shall show a crack at the surface.

3.3.2 *Resistance to wetting and drying.*

3.3.2.1 *Type I.*—The assembly shall not delaminate more than 25 percent in any adhesive bond (between plywood and decorative laminate, between plywood and backing, or in the plywood) when tested as specified in 4.3.3.2.2.1.

3.3.2.2 *Type II.*—The assembly shall not delaminate more than 25 percent in any adhesive bond (between plywood and decorative laminate, between plywood and backing, or in the plywood) when tested as specified in 4.3.3.2.2.2.

3.3.3 *Resistance of the table top assembly to high temperature (grades 1 and 2).*— There shall be no blisters, cracks, or discoloration of the decorative laminate when the assembly is tested as specified in 4.3.3.2.3.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. — At least one assembled table top with an area of at least four square feet for each combination of type and grade specified (the edging need not be applied to this sample), and a piece of decorative laminate with an area of at least two square feet for each combination of grade and color specified, shall be used to determine conformance of the material to this specification.

4.2 Inspection.—The samples shall be inspected for type of material, color, finish, defects, workmanship, and for conformity with all applicable requirements of this specification. Inspection shall also include packaging, packing, and marking for shipment.

4.3 Tests.

4.3.1 *Certification.* — When specified, the contractor shall forward to the procuring agency with each shipment of assembled table tops and decorative laminate a certified statement (1) identifying the adhesive used in the core and in attaching the faces to the core, and (2) certifying that the table top assembly conforms to the requirements of this specification.

4.3.2 *Retest and rejection.*—If the results of any test do not conform to the requirements prescribed in this specification, at the option of the purchaser, two additional sets of specimens from the same shipment may be tested, each of which shall conform to the requirements specified. If either of these two additional sets of specimens fail, the shipment of the table top assemblies may be rejected at the option of the purchaser.

4.3.3 *Test methods.*—The test specimens shall be conditioned and tested in accordance with the requirements of Federal Specification L–P–406, unless other conditions are specified in this specification or in the test method. In particular, specimens of the decorative laminate shall be conditioned for at least 48 hours at 23.1° C. (73.5° F.) and 50 percent relative humidity before test, and specimens of the table top assembly shall be kept at the same conditions at least 96 hours before test.

4.3.3.1 *Tests of the decorative laminate.*

4.3.3.1.1 *Thickness.* — The thickness shall be measured to the nearest 0.001 inch at three different locations on the specimen.

4.3.3.1.2 *Colorfastness to light.* — Three specimens shall be tested in accordance with method No. 6031.

4.3.3.1.3 *Flexural properties.* — Flexural tests shall be made on 5 specimens each from

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the lengthwise and crosswise directions of the sheet in accordance with method No. 1031. The specimens shall be tested face-down. The rate of crosshead motion shall be 0.05 inch per minute till failure. A secant flexural modulus of elasticity shall be computed as follows: A load equivalent to a stress of 10,000 pounds per square inch (P = 417bd) shall be marked on the load-deflection graph. The flexural modulus of elasticity shall be calculated from the formula:

$$E = \frac{427{,}000\ d}{Y}$$

where:

Y = deflection at stress of 10,000 pounds per square inch.

4.3.3.1.4 *Abrasion.*—Three specimens shall be tested in accordance with method No. 1091. Each specimen shall be abraded for a total of 5500 revolutions. The specimen shall be weighed after 500 and 5500 revolutions. The average weight loss per 1000 revolutions, based on the difference in weights at 500 and 5500 revolutions, shall be reported.

4.3.3.1.4.1 *Specimen for abrasion.* — The specimen shall be 4.05 inches across, either circular or square. Most decorative laminates show warp and must be held flat in the abrasion test. This may be done by holding a circular specimen flat with a collar which is standard equipment with the Taber abrasion machine, or by gluing a specimen to 0.10-inch-thick-aluminum. Another method is to hold a specimen to 0.10-inch-thick-sheet of metal with bolts and washers at the corners of the square sheet. A 1-inch diameter hole is cut in the center of these last two specimens.

4.3.3.1.5 *Water absorption.*—Three specimens shall be tested in accordance with method No. 7031.

4.3.3.1.6 *Test for resistance to chemicals.* —Three drops of each of the following aqueous solutions shall be placed on separate spots on the decorative laminate, or on separate specimens thereof; (1) trisodium phosphate, 5 percent solution; (2) phenol, 5 percent solution; (3) ammonia, 10 percent solution; (4) citric acid, 10 percent solution; (5) ethyl alcohol, 95 percent solution; and (6) sodium hydroxide, 1 percent solution. The drops shall be covered with a watch glass and permitted to stand for 24 hours. The report shall include observations as to swelling, blistering, and change in color for each of the test solutions.

4.3.3.2 *Tests of table top assembly.*

4.3.3.2.1 *Impact resistance.*—The specimen shall be a piece of the table top assembly, 6 by 12 inches. Lines, 2 inches apart, shall be drawn on the decorative laminate side of the specimen with a China marking pencil. The specimen shall be securely fastened to a level hardwood supporting surface. An 8-ounce (1.50 inches diameter) steel ball shall be dropped on the specimen onto the 10 points where the lines intersect. None of these points shall be less than 2.0 inches from the edge of the test specimen. The ball shall be dropped from a height of 24 inches (distance between the bottom of the ball and the decorative laminate). The decorative laminate shall be inspected for cracks.

4.3.3.2.2 *Resistance to alternate wetting and drying.*

4.3.3.2.2.1 *Type I.* — Three specimens 1.5 inches square shall be given 5 cycles of the following exposure:

Period hours	*Temperature ° C.*	*Condition*
4	97 — 100	Immersion in boiling water
20	60 ± 3	Less than 10 percent relative humidity in a circulating-air oven

4.3.3.2.2.2 *Type II.*—Three specimens 1.5 inches square shall be given 5 cycles of the following exposure:

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Period hours	*Temperature ° C.*	*Condition*
16	23 ± 1.1	Immersion in water
8	60 ± 3	Less than 10 percent relative humidity in a circulating-air oven

4.3.3.2.3 *Resistance to high temperature.* —Two specimens 9 inches square shall be cut from the table top assembly. A flat-bottom round aluminum vessel having a capacity of approximately one pint, Fisher Scientific Company, Cat. No. 2-542, No-Bump Solution Evaporator, 600 milliliter, with the bottom sanded flat, or equivalent, shall be filled with wax or paraffin, heated to 180° ± 1° C. (356° ± 2° F.), and then placed on the center of the test specimen for 20 minutes. The surface of the specimen shall be inspected for blisters, cracks and discoloration.

4.3.3.3 *Tests of paper phenolic laminate backing material.*

4.3.3.3.1 *Flexural strength.* — Five specimens each from the lengthwise and crosswise directions of the sheet shall be tested in accordance with method No. 1031.

4.3.3.3.2 *Water absorption.*—Three specimens shall be tested in accordance with method No. 7031.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified in the contract or purchase order, the table top assemblies shall be packaged in accordance with accepted commercial practice.

5.2 **Packing.**—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Individual table tops.* — Each table top shall be marked on the back with a designation of type and grade and the name of manufacturer.

5.3.2 *Issue packages.* — Unless otherwise specified in the contract or purchase order, issue packages shall be marked with the name of the commodity, the quantity contained therein, and the names of the contractor and manufacturer.

5.3.3 *Shipping containers.*—Unless otherwise specified in the contract or purchase order, shipping containers shall be marked with the name of the commodity, as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 **Intended use.**—The table top assemblies covered by this specification are intended for use in dining rooms and cafeterias as table tops and counters, and in kitchens as working surfaces in the preparation of food. This specification does not cover table top assemblies for use in chemical laboratories. It is recommended that type I table top assemblies be used where exposure to water is frequent, as in installations near sinks.

6.2 **Ordering data.** — Purchasers should specify required type, grade, dimensions, color and pattern, finish, and type of trim, and should exercise any desired options offered herein. (See 1.2, 3.1.2, 3.1.2.2, 4.1, 4.3.1, 5.3.2, 5.3.3, and 6.3)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If samples are required with bid, this requirement should be stated in the inquiry or invitation for bids. (See 4.1.)

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6.4 Transportation description.

6.4.1 *Transportation description applicable to these items is:*

Tops, table, wooden, covered with laminated plastics.

Carload weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload weight 24,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN–P–103—Packaging and Packing for Overseas Shipment—Boxes; Wood-Cleated, Solid-Fiberboard.
JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.
MIL–L–10547—Liner, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency, or as directed by the contracting officer.)

7.2.2 *Packing.*

7.2.2.1 *For domestic shipment and storage.*—Unless otherwise specified, table tops packaged as described in 5.1 shall be packed in wood-cleated fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Federal Specification NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631 and LLL–B–636, respectively. The gross weight of wood or wood-cleated boxes shall not exceed approximately 100 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

7.2.2.2 *For overseas shipment.*—When specified, table tops packaged as described in 5.1 shall be packed in wood-cleated solid fiber-

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board, cleated-plywood, nailed wood, wirebound, fiberboard, or fiberboard-lined wood boxes conforming to Military Specification JAN–P–103, JAN–P–105, JAN–P–106, MIL–B–107, JAN–P–108, and JAN–P–138, respectively. The containers shall be fitted with a waterproof case liner conforming to the type I, grade A or B, class 2 requirements of Military Specification MIL–L–10547 and sealed in conformance with the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed approximately 100 pounds, and of fiberboard boxes approximately 70 pounds.

7.2.3 *Marking.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.4 *Ordering data.*—Procurement documents should specify whether domestic or overseas packing and marking are required (see 7.2.2).

7.2.5 *Sampling.*—Sampling shall be conducted in accordance with Military Standard MIL–STD–105.

NAVY INTEREST: *MC* S Y

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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